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Examining the Effects of Household Chaos on Child Executive Functions: A Meta-Analysis

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Household chaos—characterized by disorganization and instability within the home—is inversely associated with child executive functioning and effortful control, although a quantitative synthesis of the findings is currently lacking. This meta-analysis incorporated 35 studies with 36 independent effect sizes including 16,480 children. Several moderators were assessed, the most fundamental involving method (i.e., informant-completed questionnaire vs. direct assessment) of assessing executive functions. The analyses revealed a significant overall effect of $r = .16$. Only measurement approach of executive functions significantly moderated this association: Informant-completed questionnaires yielded an effect of $r = .23$, as compared with direct assessment, $r = .11$; however, both effects were significant. Based on substantive and statistical considerations, questionnaire and direct assessment effects were then meta-analyzed separately. Although effect sizes proved heterogeneous in the context of questionnaires, analyses revealed no significant moderators. Within direct assessment effects, both household chaos dimensions were significantly related to child executive functions, but instability was a stronger correlate ($r = .17$) than disorganization ($r = .06$). Results highlight the need for fuller investigation of differences in construct measurement presented by questionnaire and direct assessment approaches to child executive functions. At present, it appears prudent to adopt a multimethod approach to assessment. Ultimately, a greater focus on quasi-experimental designs examining the mechanisms by which events destabilize child executive functions is essential to a rigorous understanding of the relation between environmental features and child cognitive function.

Public Significance Statement

This meta-analysis provides evidence of the direct negative effect of household chaos on child executive functioning. It also contributes to the ongoing discussion regarding strategies of measuring executive functioning in children, with a multimethod, longitudinal approach potentially being most robust. Further, it highlights the exacerbating effects of familial dynamics on the impact of household chaos on child executive functioning, and in particular the importance of mitigating levels of instability in their lives.

Keywords: children, direct assessment, executive functions, household chaos, informant-completed questionnaires

Household chaos is described as “systems of frenetic activity, lack of structure, unpredictability in everyday activities, and high levels of ambient stimulation” (Bronfenbrenner & Evans, 2000, p. 121). It is

characterized in the literature by two broad features: disorganization and instability. Indices of disorganization include clutter, ambient noise, crowding and lack of structure (Matheny, Wachs, Ludwig, &

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Data are available in the Open Science Framework (OSF) repository via the following link: <https://osf.io/2gxxk3>.

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Phillips, 1995; Sameroff, 2010). Instability refers to frequent changes in residence or residents (e.g., primary or secondary caregivers) and a lack of or unpredictable routines (Wachs & Evans, 2010). According to the bioecological theory, child development is shaped by the increasingly complex reciprocal interactions (i.e., proximal processes) that the child has with elements of their immediate environment (Bronfenbrenner & Morris, 1998, 2006). These proximal processes have the greatest influence on child development when they are consistent, progressively complex, reciprocal, and occur over extended periods of time (Bronfenbrenner & Evans, 2000; Bronfenbrenner & Morris, 1998). Household chaos can disrupt the development and sustainability of these experiences of the child (e.g., relationships, activities, routines; Bronfenbrenner & Evans, 2000; Evans, Gonnella, Marcynyszyn, Gentile, & Salpekar, 2005). This is particularly detrimental for younger children, where one's immediate built environment is important for early establishment of core capacities (Dunn, 2012; Oliver, Dunn, Kohen, & Hertzman, 2007). As such, the literature has demonstrated linkages between household chaos and numerous child and adolescent outcomes including deficits in socio-emotional development (Evans et al., 2005; Fiese & Winter, 2010), poor language and early literacy development (Johnson & Martin, 2008; Vernon-Feagans et al., 2012), lower intelligence levels (Deater-Deckard et al., 2009), deficits in academic achievement (Garrett-Peters et al., 2016), and conduct problems and callous-unemotional behaviors (Mills-Koonce, Willoughby, Garrett-Peters, Wagner, & Vernon-Feagans, 2016). Relevant to this review, higher household chaos has also been linked to deficits in executive functions (Brieant, Holmes, Deater-Deckard, King-Casas, & Kim-Spoon, 2017; Brown, Ackerman, & Moore, 2013; Hughes & Ensor, 2009), with many, but not all studies showing significant associations. Although findings are mostly consistent, there is variation across studies, with few identifying reliable moderators. No study to date has formally integrated these findings and examined the interaction between household chaos and the measurement approach of executive functions (i.e., informant-completed questionnaires vs. direct assessment). Nor have studies systematically evaluated the influence of other potential moderators (e.g., disorganization vs. instability, child gender, age, race/ethnicity). The present meta-analysis assesses the strength of association between household chaos and child executive functions and examines moderators of this association.

Child Executive Functions

Executive function is an umbrella term for higher order cognitive processes subsumed by the prefrontal cortex and is critical to flexibility, planning, and goal directed behavior (Diamond, 2013). Based on theoretical and empirical considerations (Diamond, 2013; Tiego, Bellgrove, Whittle, Pantelis, & Testa, 2020; Zhou, Chen, & Main, 2012) and given that many studies assess varied constructs in children, the current meta-analysis includes inhibition, cognitive flexibility, working memory, effortful control, and attention as measures of child executive function. *Inhibition* refers to the ability to suppress automatic or dominant responses via control over attention, thoughts, emotions, and actions (Diamond, 2013). *Cognitive flexibility* or *set-shifting* describes the ability to transition between modes of mental operations, tasks, or cognitive rules (Miyake et al., 2000; Willoughby, Wirth, & Blair, 2012). *Working memory* refers to the active retention, manipulation, and processing of information (Willoughby et al., 2012). *Effortful*

control is a component of temperament and refers to the ability to regulate one's behavior via inhibitory control, planning actions, detecting and correcting errors, resolving conflict, as well as shifting attention (Diamond, 2013; Rothbart, Sheese, & Posner, 2007). Lastly, *attention* comprises three interlinked neural networks (alerting, orienting and executive attention) and is also inherent to effortful control (Rothbart & Rueda, 2005; Simonds, Kieras, Rueda, & Rothbart, 2007). Inclusion of this set of executive functions is important to a robust assessment of the effects of household chaos documented in the literature.

Household Chaos and Child Executive Functions

Rapid development of executive functions in childhood makes exposure to contextual risk factors such as household chaos particularly threatening. Indeed, direct associations have been made between elevated levels of household chaos and poor performance in tasks related to core executive functions (inhibitory control, cognitive flexibility, working memory; Hughes & Ensor, 2009; Hur, Buettner, & Jeon, 2015), as well as attention (Brown, Weatherholt, & Burns, 2010) and effortful control (Chen, Deater-Deckard, & Bell, 2014).

Direct exposure to household chaos may prompt children to develop strategies to filter out the excess stimulation which may, in turn, also prevent them from benefiting from positive environmental influences (Evans, Kliewer, & Martin, 1991). Alternatively, young children may not have the capacity to filter out irrelevant stimuli which may become overstimulating (Lillard, Drell, Richey, Boguszewski, & Smith, 2015; Wachs & Evans, 2010) and threaten the development of their regulatory systems. Finally, it is also possible that cumulative chaos may interfere with a child's competency level, increasing a sense of helplessness to control the events within their environment (Evans & Stecker, 2004). The mechanism of effect has not been fully elucidated yet.

However, of consideration is the further differentiation of household chaos and executive function by definitional variations and assessment measures. As mentioned above, household chaos is operationalized as disorganization and instability. In addition, there is variation in measurement approach of executive function: using either questionnaires (e.g., Behavior Rating Inventory of Executive Function [BRIEF]; Gioia, Isquith, Guy, & Kenworthy, 2000) or direct assessments (e.g., Dimensional Change Card Sort [DCCS]; Zelazo, 2006). Consequently, although associations between household chaos and child executive functioning have been demonstrated, effect sizes vary widely. As such, the definition of household chaos and the measurement of executive functions have important implications for this meta-analysis and, therefore, served as moderators in examining this association. Other relevant moderators are also assessed as described below (i.e., child age and sex, socioeconomic status [SES] indices, and temporal characteristics).

Effect of Household Chaos Dimension on Child Executive Function

The conceptualization of the two dimensions of household chaos, disorganization, and instability is broad, and not all factors are mutually exclusive (e.g., household routines, crowding). However, few studies have examined the potential differential effects of

each dimension of household chaos on child outcomes (Wachs & Evans, 2010) including executive functions. Differential effects may be rooted in the frequency with which a child is exposed to elements of each dimension. Children may be more chronically exposed to elements of household disorganization (e.g., ambient noise and clutter) compared with elements of household instability (e.g., residential mobility, changes in caregivers). Further, the persistent exposure to disorganization may result in a child becoming habituated to their circumstances and adverse effects may emerge over time. In contrast, the unpredictability of elements of instability may be more threatening to the consistency of the child's proximal processes, thus leading to more salient and prominent difficulties in regulatory processes and behavior. Empirically, household disorganization negatively influences inhibitory control, attention shifting, and working memory (Berry et al., 2016). Greater household instability (e.g., caregiver structure changes, residential mobility, job/income loss, familial death) has been associated with more difficulties with effortful control in children from ages four to six years (Sturge-Apple, Davies, Cicchetti, Hentges, & Coe, 2017). Similarly, a lack of routine experienced by preschool children has been linked to poor performance on delayed gratification tasks at kindergarten-age (Martin, Razza, & Brooks-Gunn, 2012). Correlations between disorganization and instability are rarely examined, with one large, longitudinal study, the Family Life Project, reporting correlations ranging from $r = .38$ to $.42$ (Berry et al., 2016; Garrett-Peters et al., 2016; Vernon-Feagans, Willoughby, & Garrett-Peters, 2016). However, each dimension also accounts for unique (nonshared) variance. Given the theoretical importance of these two dimensions, the potential differential effects of instability and disorganization were examined as a moderator.

Measurement of Executive Functions

Executive functions of children can be measured via empirically validated questionnaires and/or direct assessments. Questionnaires, such as the BRIEF (Gioia et al., 2000), are typically completed by parents or teachers and are related to children's abilities in navigating home and school contexts. These questionnaires aim to assess a number of executive function domains including working memory, inhibition, set-shifting and emotional control. Direct assessments, for example the DCCS (Zelazo, 2006), are typically completed by the child within highly standardized conditions with each task designed to measure one or more specific executive function. Low correlations between direct assessment and questionnaires are common (Anderson, Anderson, Northam, Jacobs, & Mikiewicz, 2002; Mahone et al., 2002), suggesting that their shared variance is limited. A systematic review of 13 studies focusing on child and adolescent populations (with the exception of one study looking at young adults), assessed the BRIEF against direct assessments (Toplak, West, & Stanovich, 2013). The mean correlation was $r = .15$, with only 19% significant. A more recent study provided similar results; aside from working memory, the scales on the BRIEF did not significantly correlate with their corresponding direct assessment tasks (Krivitzky, Bosenbark, Ichord, Jastrzab, & Billingshurst, 2019). These low correlations extend to other child executive function questionnaires (range across scales, $r = .01$ to $-.15$, *ns*; Emslie, Wilson, Burden, Nimmo-Smith, & Wilson, 2003; Roy, Allain, Roulin,

Fournet, & Le Gall, 2015). Given these findings, it is possible that questionnaires and direct assessments measure distinct domains of executive functioning. Although direct assessments may detect the processing efficiency of executive functions in children during highly structured tasks, questionnaires may provide a more ecologically valid perspective in measuring rational goal pursuit (Toplak et al., 2013). Furthermore, these approaches may also differ in their association with other neurodevelopmental processes. For example, while both questionnaires and direct assessment predicted reading and mathematical ability in school-age children; they differed in their capacity to predict attention and motor ability (Ten Eycke & Dewey, 2016). Questionnaires significantly predicted measures of attention, such that lower executive function ability predicted higher attention problems, whereas the direct assessment significantly predicted measures of motor ability, such that higher executive function ability predicted better motor ability.

These findings are further supported by research suggesting that questionnaires and direct assessments may have differing underlying neurobiological substrates. One study showed that, for working memory, there was no association between questionnaires and lobar volumes, however, direct assessments were negatively correlated with the volume of the right hippocampal lobe (Faridi et al., 2015). Additionally, negative correlations were found between questionnaires and bilateral cortical thickness of the posterior parahippocampal gyrus (PG), whereas no such associations were found with tests of direct assessment. Collectively these findings highlight the importance of analyzing the two methodological approaches separately.

Development of Executive Functions

Basic components of executive functions (e.g., delayed response or holding information in one's mind) emerge within the first three years of life, with attention underlying this stage of development (Garon, Bryson, & Smith, 2008; Garon, Smith, & Bryson, 2014). Maturation toward more complex executive functions (e.g., inhibition, working memory, set-shifting) occurs in a stepwise fashion with key developmental periods occurring in early childhood (ages three to five), middle childhood (ages seven to nine), and extending throughout adolescence into early adulthood (Anderson, 2002; Carlson, 2005). Neuroimaging studies have corroborated neuropsychological changes underlying this stepwise development. Maturation of cortical structures (e.g., gray matter loss; pruning) are aligned with major developmental milestones (e.g., cognitive, functional) across childhood and adolescence (Gogtay et al., 2004) through to early adulthood (Diamond, 2006; Pechtel & Pizzagalli, 2011). This prolonged development leaves executive functions vulnerable to the detrimental effects of early adversity, which may last beyond cessation of the adverse factors (Pechtel & Pizzagalli, 2011). Given these developmental and neural trajectories underlying executive functions across childhood through to adulthood, child age was evaluated as a moderator.

Sociodemographic Effects on Executive Functions

Sociodemographic factors (e.g., income, parental education, racial/ethnic status) are related to individual differences in executive functions and have been extensively explored in the literature with

consistent trends toward low SES and difficulties in child executive functioning (Holochwost et al., 2016; Lawson, Hook, & Farah, 2018; Raver, Blair, Willoughby, & The Family Life Project Key Investigators, 2013). Although household chaos can occur within higher SES homes, many indicators of poverty such as low-income housing, nontraditional work schedules, and inaccessible child-care, put families at greater risk of experiencing household chaos (Evans, Eckenrode, & Marcynyszyn, 2010; Lichter & Wethington, 2010). In fact, early, cumulative household chaos mediates the association between indicators of poverty and child (Dumas et al., 2005) and adolescent (Evans et al., 2005) socioemotional outcomes. As such, household chaos is an important mechanism through which poverty can exert its effects on children above and beyond other demographic (e.g., parental education), familial (e.g., parenting), and early cognitive abilities (Garrett-Peters et al., 2016). Thus, we considered the moderating effect of two proxies of SES (e.g., parental education, racial/ethnic status) on the association between household chaos and child executive functions.

Sex Differences in Executive Functions

Although development of executive functions is similar in males and females, some differences have been noted in a few studies such as female children outperforming males in measures of self-regulation (i.e., attentional, inhibitory, and effortful control; Moilanen, Shaw, Dishion, Gardner, & Wilson, 2010; Raikes, Robinson, Bradley, Raikes, & Ayoub, 2007). One possible explanation is that the development of executive functions in male children may be slower than that of female children, particularly in early childhood (Gunzenhauser, Saalbach, & Von Suchodoletz, 2017; Matthews, Ponitz, & Morrison, 2009; Ponitz, McClelland, Matthews, & Morrison, 2009). Thus, it may be that male children, experiencing slower developmental gains in their executive function skills, are more vulnerable to contextual threats. Despite these potential developmental trajectories in males and females, a recent review examining sex differences in the performance of children and adolescents (among other human and animal models) on executive function tasks reported mixed findings (Grissom & Reyes, 2019); suggesting that this warrants further investigation. For the purposes of this meta-analysis, child sex was examined as a potential moderator influencing the association between chaos and executive function.

Temporal Characteristics

The further the assessments are from one another, the smaller their correlation may be, owing to time-related intervening factors (e.g., Atkinson et al., 2000). Thus, temporal distance between household chaos and child executive function assessments was assessed as a moderator.

The Current Study

The aim of the current study is threefold: (a) to examine the magnitude of effect between household chaos and executive functions in children and adolescents, (b) to examine potential moderators that influence the association between household chaos and executive functions, and (c) to identify gaps within the household chaos and executive functions literature.

Method

Operational Definitions

The aim of the current meta-analysis was to synthesize the literature examining the association between household chaos and child executive functioning. Studies which measured one or both dimensions of household chaos (disorganization and instability) were included. Elements of disorganization included crowding/density, clutter, lack of structure and ambient noise (e.g., neighborhood or TV noise; Sameroff, 2010), while instability referred to changes in residences, changes in primary or secondary caregivers, and a lack of or unpredictable routines (Wachs & Evans, 2010). Studies that incorporated household chaos as a component of a larger construct of socioeconomic status were considered if they presented data on household chaos distinct from the overall construct. To account for varied executive function assessment methods, both informant-completed questionnaires and direct assessment measures were included. Direct assessment tasks that measured core executive functions included: working memory (e.g., Digit Span; Wechsler, 2014), cognitive flexibility (e.g., DCCS; Zelazo, 2006), and inhibition (e.g., Day-Night Stroop; Gerstadt, Hong, & Diamond, 1994). Measures of attention and effortful control were also included. Questionnaires, such as the BRIEF (Gioia et al., 2000), were also included.

Literature Review

A comprehensive search was conducted of the following databases: Medline, EMBASE, PsycINFO, ERIC, PubMed, and ProQuest Dissertations & Theses A&I covering all studies from inception of the database to April 2020. Relevant key search terms were categorized under three primary variables. To capture data for household chaos: *home**, *house**, *famil**, *residen** and *environment** were used in conjunction with: *chaos*, *disorgan**, *disorder**, *instabilit**, *crowd**, *dysfunc**, and *unpredict**. Executive function was examined via: *executive control**, *executive process**, *effortful control*, *inhibit**, *impuls**, *executive attent**, *attent**, *working memory*, *set-shift**, and *cognitive flexibilit**. Participants were narrowed to children and adolescents with the following terms: *child**, *girl**, *boy**, *infant**, *toddler**, *preschool**, *young person**, *minor**, *teen**, *adolescen**, *youth**, *school-aged*, and *early childhood**. Bibliographies from relevant reviews and book chapters were manually searched for additional citations.

Inclusion Criteria

The following inclusion criteria were applied: (a) sample consisted of children and/or adolescents (aged 2 to 17 years); (b) studies included a measure of one or more dimensions of household chaos (e.g., disorganization and/or instability) distinct from other related constructs (e.g., indices of SES); (c) studies included at least one measure of executive function as a correlate of household chaos, including informant-completed questionnaires and direct assessments; and (d) articles were written in the English language.

Selection of Studies

Figure 1 shows the process of selection and exclusion of articles. The initial search yielded 13,400 nonduplicate studies. After titles

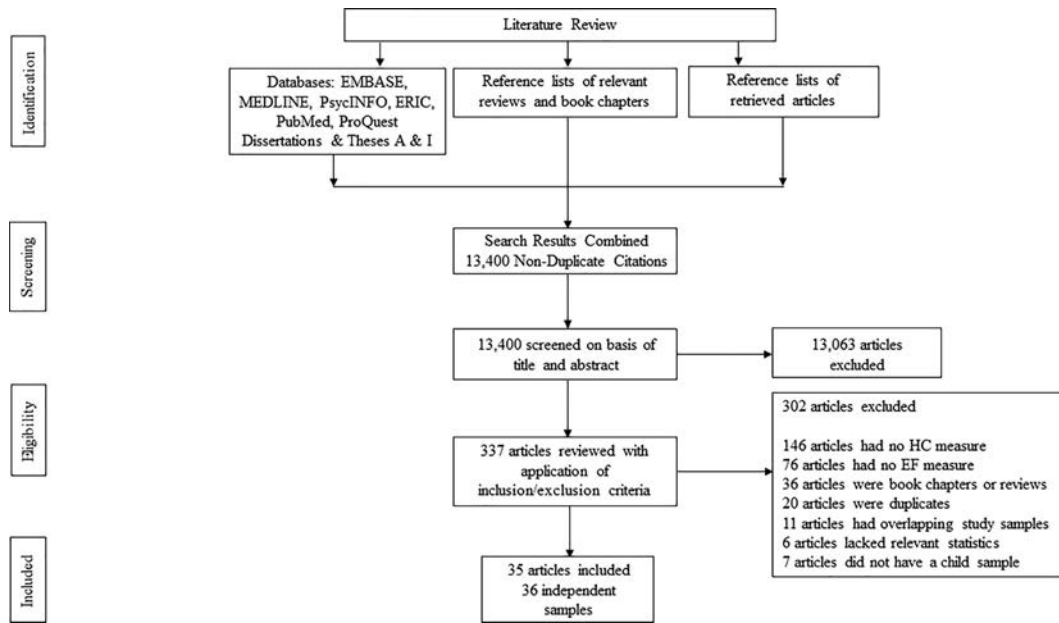


Figure 1. Flowchart of study selection process for meta-analysis on household chaos and child executive function literature.

and abstracts were screened to determine relevance, 337 articles remained. Full text review of these articles resulted in elimination of an additional 285 articles. The remaining 52 articles underwent data extraction during which an additional 11 were removed because of overlapping study samples; in such cases, the study with the largest sample size was selected. Six additional articles were not included in the main analysis due to inclusion of covariates or lack of relevant statistics. Two independent reviewers (Krysta Andrews and Madeleine Harris) conducted screens of articles at each stage. Disagreements between reviewers were resolved via an in-depth discussion and thorough review of relevant sections of the articles in question. Interrater reliability was established by coding 10% articles at the title and abstract screen, 20% at the full text screen and 100% at data extraction. Percent agreement was greater than 80% at each stage ($kappa = .77$; $p < .0001$). To address insufficient moderator data, 37 authors were contacted via e-mail. Of those contacted, 11 authors provided the requested information, two authors provided partial data requested, nine authors responded and did not have the requested information (e.g., not measured, no access to data), one declined to provide requested information, and no response was received from the remaining 14 authors. Final consensus between reviewers resulted in 35 articles (36 data sets) included in the meta-analysis.

Coding

A single Pearson correlation (r) between measures of household chaos and executive functions was entered for each sample. Valence of each correlation was adjusted to account for the direction of effect. Effect sizes indicating a negative correlation between household chaos and executive functions were considered positive (i.e., higher household chaos linked to lower executive function performance), representing the expected direction of association.

One study provided correlational values via a bar graph (Hughes & Ensor, 2009); however, the value was extracted using the Web Plot Digitizer program (Rohatgi, 2018). Six studies were excluded from the meta-analysis ($k = 36$), but in each case an r of 0.0 was imputed (Rosenthal, 1995) and entered into a sensitivity analysis ($k = 42$) to assess robustness of effect size parameters.

Two independent reviewers coded the moderator information for each article including: (a) approach to measurement of executive function; (b) household chaos dimension (disorganization vs. instability); (c) sex composition of sample (percentage of female children); (d) mean child age in years at time that executive function was measured; (e) race/ethnicity (percentage of minorities in sample); (f) parental education (percentage of parents with less than or equal to a high school diploma or General Educational Development) (GED); and (g) months between assessments of household chaos and child executive function.

For the purposes of this review, studies that used the Confusion, Hubbub, and Order scale (CHAOS) were coded as disorganization. The majority of the items measure elements of disorganization or environmental confusion (e.g., ambient noise, clutter, crowding) as per the scale's original development, with the exception of one item which characterizes instability; a question related to the level of routines in the home (Matheny et al., 1995). Some studies opt to add measures of instability to supplement the CHAOS scale (e.g., Family Instability Questionnaire [FIQ]; Ackerman, Kogos, Youngstrom, Schoff, & Izard, 1999; Forman & Davies, 2003) and to ensure a more comprehensive assessment of household chaos. Sample items of the CHAOS Scale include: "It's a real zoo in our home"; "We can usually find things when we need them"; and "You cannot hear yourself think in our home." For the current meta-analysis, studies that measured elements such as clutter, ambient noise, crowding, renovations, or ongoing repair were

categorized under the *disorganization* dimension (Matheny et al., 1995; Sameroff, 2010). Studies measuring elements such as changes in residence, residents (e.g., primary or secondary caregivers), changes in job/income, or a lack of or unpredictable routines were categorized under the *instability* dimension (Wachs & Evans, 2010). Two studies composited disorganization and instability measures (Brown et al., 2013; Evans et al., 2005); therefore, data regarding household chaos dimension for these two studies were considered missing and not included in the moderation analysis.

For child age, where multiple measures were administered across time (both within and across studies), the age at which assessments of household chaos and executive functioning were most proximal was chosen. One study (Burney, 2010) reported an age range, and using a pre-established method (Wan, Wang, Liu, & Tong, 2014), the sample mean age and standard deviation were estimated.

Interrater Reliability: Effect Sizes and Moderators

Interrater reliability analyses were conducted to establish consistency in coding of effect sizes and moderators by two independent raters (Krysta Andrews and Madeleine Harris). For effect sizes and sample sizes, interrater reliability was ICC (3, 1) = .81 and ICC (3, 1) = .99, respectively. Reliability for moderator data were as follows: (a) approach to measurement of executive function ($kappa = .92$; $p < .0001$), (b) household chaos dimension ($kappa = 1.0$; $p < .0001$), (c) sex composition (ICC [3, 1] = 1.0), (d) child age (ICC [3, 1] = .87), (e) race/ethnicity composition (ICC [3, 1] = .99), (f) parental education (ICC [3, 1] = .86), and (g) time between assessments of household chaos and child executive function (ICC [3, 1] = .82).

Analytical Strategy

The Comprehensive Meta-Analysis (CMA, Version 3.0) software program (Borenstein, Hedges, Higgins, & Rothstein, 2013) was used to calculate effects. Summary effect sizes were calculated as correlation coefficients (r) within a random effects model. Effect sizes were inversely weighted according to their variance to minimize upward-biased effects of smaller studies and thus produce more accurate overall effect size estimates (Hedges & Olkin, 1985). The CMA software was also used to compute within study effect size composites based on multiple outcomes, tests of heterogeneity, proportion of between-sample variance, moderation, and publication bias.

Notably, three meta-analyses were conducted in the current study. The initial overall meta-analysis included all data sets ($k = 36$), regardless of measurement approach. In this analysis, five studies used both questionnaires and direct assessment to assess executive functioning (Brown et al., 2010; Gaertner, 2012; Lanza & Drabick, 2011; Lecheile, Spinrad, Xu, Lopez, & Eisenberg, 2020; Rea-Sandin, 2018). Five effect sizes pertaining to informant-completed questionnaires were removed to ensure sample independence in the comparison of questionnaire and direct assessment effects and to ensure equal representation of effect sizes across measurement approach. Subsequent to this initial overall analysis, independent analyses were conducted on each of the informant-completed questionnaire and direct assessment data sets. For these

separate analyses, the five effect sizes previously removed were reinserted into the questionnaire analysis ensuring maximal use of the data and augmented moderation analysis for factors other than measurement approach. The total number of samples in the questionnaire and the direct assessment analyses was 20 and 21, respectively.

Statistical Independence

For each study, the effect size (r) was calculated as the correlation between household chaos and executive functions. In most instances, data sets only had a single measure of executive function ($k = 16$); in others, authors provided a composite of executive function measures ($k = 3$). Some data sets provided correlations between household chaos and each subscale of a single measure ($k = 3$) or multiple executive functions tasks ($k = 9$). These cases required the extraction of the correlation between household chaos and each subscale or executive function measure as well as the intercorrelations among subscales or executive function measures (provided in the studies) to compute one effect size per dataset via procedures previously outlined (Borenstein, Hedges, Higgins, & Rothstein, 2009). Similarly, although most data sets provided a single measure of household chaos, four had multiple measures of either household disorganization or instability, each with a correlate for executive function (Ferretti & Bub, 2017; Squires, 2017; Vrijhof, van der Voort, van IJendoorn, & Euser, 2018; Warren & Barnett, 2020). In this case, a single effect size was generated by statistically combining the correlates via the aforementioned strategy (Borenstein et al., 2009). Finally, one dataset (Berry et al., 2016) had separate measures for disorganization and instability, each with a correlate for executive function. An a priori decision was made to select the correlation between instability and child executive functions to ensure comparability of cell sizes for this study.

Tests of Heterogeneity and True Between-Sample Variance

Heterogeneity of the overall set of effect sizes was tested via computation of the Q and the I^2 statistics (Borenstein et al., 2009). The Q statistic provides a ratio of observed variation to within study variation of the effect sizes (Borenstein et al., 2009). The I^2 statistic is an estimate of the proportion of variance that is attributable to real population differences (Borenstein et al., 2009; Higgins, Thompson, Deeks, & Altman, 2003).

Moderator Analyses

To assess for moderation, Q -statistics were calculated in the case of categorical variables (i.e., approach to measurement of executive functions, household chaos dimension). For dimensional moderators (i.e., sample sex composition, mean child age, racial/ethnic sample composition, parental education, and temporal distance between household chaos and executive function measurements), metaregression was employed, using a method of moments approach. This is a conservative method which makes no assumptions regarding distribution properties. As per a previously established convention (Borenstein et al., 2009), analyses involving categorical variables were only conducted for moderators with five

or more observations per cell so as to ensure viable estimates of between-sample variance.

Publication Bias

Potential publication bias was assessed using funnel plots of the effect sizes against their standard error. Additionally, Duval and Tweedie's (Duval & Tweedie, 2000a, 2000b) "trim and fill" method, based on the symmetry of the funnel plots, was used to impute the possible existence and impact of unpublished studies (Borenstein et al., 2009). This included derivation of an adjusted overall effect size and confidence interval.

Results

Study Characteristics

A total of 35 articles, including 36 independent samples and 16,480 participants, are represented in the analysis. Of the 36 samples, 30 were presented in published articles and six in unpub-

lished dissertations. Articles were published between 2005 and 2020. Studies were predominantly conducted in the United States (32), with others conducted in the Netherlands, the United Kingdom, Germany, and Australia. Study characteristics and effect sizes are provided in Tables 1 and 2, respectively.

Association Between Household Chaos and Child Executive Functions: Overall Analysis

Figure 2 shows sample effect sizes in a forest plot. Majority of correlations were positive, ranging from -0.01 to 0.57 . The weighted, mean association corresponds to $r = .16$, 95% CI $[-.12, .20]$. Analyses revealed significant heterogeneity among effects, $Q(35) = 363.11$, $p < .001$. As per the I^2 index, 90.36% of this variance is due to true between-sample differences. Only measurement approach to executive functions significantly moderated the association between household chaos and child executive functions. The association was significantly stronger ($Q[1] = 7.87$, $p < .01$) when assessed with informant-completed questionnaires, $r = .23$, 95% CI $[-.16, .29]$, $k = 15$, as compared with direct assess-

Table 1
Overview of Studies Included in Meta-Analysis

Study authors	N (analytic sample)	Year	Location of study	Sex composition (% female)	Racial/Ethnic composition of sample (% minorities)	Child age (years)	Parental education
Berry et al., 2016	1,235	2016	United States	50	73	4.0	NR
Brient et al., 2017	157	2017	United States	47	20	14.1	35
Brown et al., 2010	123	2010	United States	54	90	5.6	52
Brown et al., 2013	120	2013	United States	51	90	4.2	44
Burney, 2010	190	2010	United States	50	26	4.5	10
Chen et al., 2014	149	2014	United States	50	22	4.8	31
Crespo et al., 2019	85	2019	United States	46	98	3.0	62
Dumas et al., 2005	106	2005	United States	50	14	5.7	NR
Evans et al., 2005	223	2005	United States	48	6.0	13.1	NR
Ferretti & Bub, 2017	3,250	2017	United States	50	36	5.7	NR
Gaertner, 2012	230	2012	United States	44	15	2.5	17
Gould et al., 2018	1,040	2018	Australia	52	NR	9.4	NR
Gunzenhauser et al., 2017	263	2017	Germany	51	NR	8.6	NR
Hardaway et al., 2012	659	2012	United States	49	50	3.0	41
Hughes & Ensor, 2009	125	2009	United Kingdom	38	4.0	4.0	72
Hur et al., 2015	444	2015	United States	47	39	4.6	NR
Kraft et al., 2014	69	2014	United States	30	NR	3.6	NR
Lanza & Drabick, 2011	87	2011	United States	49	100	7.8	77
Lecheile et al., 2020	230	2020	United States	44	15	2.5	21
Lemery-Chalfant et al., 2013	807	2013	United States	49	10	7.9	NR
Maguire & Schneider, 2019	90	2019	United States	60	22	10.9	37
Micalizzi et al., 2019	574	2019	United States	51	10	4.1	18
Peviani et al., 2019	167	2019	United States	47	21	17.0	35
Rea-Sandin, 2018	416	2018	United States	50	41	8.4	10
Riley et al., 2020	132	2020	United States	47	49	1.9	39
Schmitt et al., 2015	359	2015	United States	49	51	4.5	NR
Squires, 2017	966	2017	United States	49	55	2.5	NR
St. John & Tarullo, 2019	121	2019	United States	58	57	5.0	11
Sturge-Apple et al., 2017 Study 1	194	2016	United States	56	54	4.1	23
Sturge-Apple et al., 2017 Study 2	201	2016	United States	44	77	4.0	30
Taylor & Lopez, 2005	200	2005	United States	52	100	14.7	NR
Valiente et al., 2007	188	2007	United States	54	75	9.6	50
Vincent, 2012	309	2012	United States	NR	8.0	11.0	12
Vrijhof et al., 2018	418	2018	Netherlands	51	NR	3.9	NR
Warren & Barnett, 2020	1969	2020	United States	46	77	3.0	60
Zhao, 2013	154	2013	United States	49	100	8.0	0.1

Note. NR = not reported; Racial/ethnic composition of sample = percentage of minorities in sample; Parental education = percentage parents with less than or equal to high school diploma/GED in the sample.

Table 2
Effect Size Characteristics

Study	Pearson's <i>r</i>	HC measure	EF measure	Measurement approach of EF
Berry et al., 2016	−0.4 −0.27	^a Indices of disorganization ^a Indices of instability	Silly Sounds Stroop Animal Go/No-Go Spatial Conflict Arrows Working Memory Span Pick the Picture Game Something's the Same Game	DA DA DA DA DA DA
Brieant et al., 2017	−0.13	CHAOS Scale	Stanford-Binet Memory for Digits Wisconsin Card Sorting Test Multi-source Interference Task	DA DA DA
Brown et al., 2010	−0.03 −0.16 −0.11 −0.27 −0.37	CHAOS Scale	Spatial Orientation of Attention Alerting Task Executive Attention Task Child Behavior Questionnaire ADHD Index	DA DA DA ICQ ICQ
Brown et al., 2013	−0.35	CHAOS Scale Residential Instability (Number of changes in where and/or with whom children had lived since their birth)	Day-Night Stroop Task Peg Tapping Task	DA DA
Burney, 2010	−0.17	CHAOS Scale	Bear/Dragon Task	DA
Chen et al., 2014	−0.24	CHAOS Scale	Child Behavior Questionnaire	ICQ
Crespo et al., 2019	−0.07	CHAOS Scale	Child Behavior Questionnaire	ICQ
Dumas et al., 2005	−0.42	CHAOS Scale	Delay Task Child Behavior Questionnaire: Attentional Focusing Subscale Child Behavior Questionnaire: Impulsivity Subscale	DA ICQ ICQ
Evans et al., 2005	−0.27 −0.17	CHAOS Scale Family Rituals Questionnaire Family Routines Inventory	Children's Self-Control Scale	ICQ ICQ
Ferretti & Bub, 2017	−0.01 −0.05 −0.05 −0.12 −0.06 −0.05 −0.08	Bedtime Routine Dinner Routine Eat at a Regular Time Reading Routine Tell Stories Routine Singing Songs Routine CHAOS Scale	Items were drawn from the Preschool and Kindergarten Behavior Scales–Second Edition and the Social Skills Rating System: Hyperactivity/inattention scale	ICQ
Gaertner, 2012	−0.07 0.05 −0.31 0.03	CHAOS Scale	Dinky Toys Gift Delay Rabbit & Turtle Task Early Childhood Behavior Questionnaire Infant Behavior Record	DA DA DA ICQ ICQ
Gould et al., 2018	−0.23 −0.30	CHAOS Scale	Strengths and Weaknesses of ADHD symptoms and Normal Behavior subscales: Inattention Subscale Strengths and Weaknesses of ADHD symptoms and Normal Behavior subscales: Hyperactive/Impulsive	ICQ ICQ
Gunzenhauser et al., 2017	0.04 0.13	CHAOS Scale	Working Memory: Corsi Block, Object Span, Digit Span Backwards Inhibitory Control/Cognitive Flexibility: Stroop Test, Go/No-Go	DA DA
Hardaway et al., 2012	−0.31 −0.02	CHAOS Scale	Child Behavior Questionnaire Behavioral Attention Styles	ICQ ICQ
Hughes & Ensor, 2009	−0.21	CHAOS Scale	Baby Stroop Task Trucks task, Beads Task Tower of London Task Head-Toe-Knee-Shoulder task	DA DA DA DA
Hur et al., 2015	−0.07	CHAOS Scale	Child Behavior Questionnaire-Short Form	ICQ
Kraft et al., 2014	−0.57	CHAOS Scale	Intra/Extra Dimensional (IED) Set Shift subtest (Cambridge Neuropsychological Test Automated Battery (CANTAB))	DA
Lanza & Drabick, 2011	−0.17	Child Puppet Interview	Child Symptom Inventory	ICQ
Lecheile et al., 2020	0.09 −0.07 −0.31	CHAOS Scale	Delay Paradigm Early Childhood Behavior Questionnaire	DA ICQ ICQ
Lemery-Chalfant et al., 2013	−0.32	CHAOS Scale	Child Behavior Questionnaire	ICQ

(table continues)

Table 2 (continued)

Study	Pearson's <i>r</i>	HC measure	EF measure	Measurement approach of EF
Maguire & Schneider, 2019	0.09	CHAOS Scale	Digit Span	DA
Micalizzi et al., 2019	−0.05	CHAOS Scale	NIH Toolbox Early Childhood Cognitive Battery: Flanker Inhibitory Control and Attention Test	DA
	−0.10		NIH Toolbox Early Childhood Cognitive Battery: Dimensional Change Card Sort	DA
Peviani et al., 2019	−0.27	CHAOS Scale	Delay Discounting Task	DA
Rea-Sandin, 2018	−0.06	CHAOS Scale	Continuous Performance Task	DA
	−0.11		NIH Toolbox Early Childhood Cognitive Battery: Flanker Inhibitory Control and Attention Test	DA
	−0.01		Digit Span Backward	DA
	−0.09		Temperament in Middle Childhood Questionnaire—Attentional Focusing	ICQ
	−0.26		Temperament in Middle Childhood Questionnaire—Inhibitory Control	ICQ
Riley et al., 2020	0.06	CHAOS Scale	Snack Delay Task	DA
Schmitt et al., 2015	−0.17	Residential Mobility Question (“How many moves has your family experienced in the past 5 years?”)	Day-Night Stroop Task	DA
Squires, 2017	−0.08	Maternal Marital Transitions	Leiter-R Examiner Rating Scales - Attention Regulation Subscale	ICQ
	−0.04	Maternal Employment Transitions		
	0.002	Residential Mobility		
St. John & Tarullo, 2019	−0.12	CHAOS Scale	Change Detection Task	DA
	0.02		Go/No-Go: Go Trials	DA
	−0.08		Go/No-Go: No/Go Trials	DA
Sturge-Apple et al., 2017 Study 1	−0.19	Family Instability Questionnaire	Peg Tapping Task	DA
	−0.06		Reward Dominance Task	DA
Sturge-Apple et al., 2017 Study 2	−0.28	Family Instability Questionnaire	Mischel's Delay of Gratification Task	DA
Taylor & Lopez, 2005	−0.21	Family Routines Inventory	School Engagement Attention Scale	ICQ
Valiente et al., 2007	−0.33	CHAOS Scale	Early Adolescent Temperament Questionnaire	ICQ
Vincent, 2012	−0.09	CHAOS Scale	Strengths and Weaknesses of ADHD symptoms and Normal Behavior subscales: Inattention Subscale (child report)	ICQ
	−0.08		Strengths and Weaknesses of ADHD symptoms and Normal Behavior subscales: Hyperactive/Impulsive (child report)	ICQ
Vrijhof et al., 2018	−0.27	CHAOS Scale	Child Behavior Questionnaire	ICQ
	−0.09	Noise (during week)		
	−0.10	Noise (during weekend)		
	−0.03	Crowding		
Warren & Barnett, 2020	0.01	Unpredictable Employment	Walk a Line Behavioral Task	DA
	0.02	Unpredictable Finances		
	0.04	Parental Unpredictability		
	−0.04	Unpredictable Daily Routines		
Zhao, 2013	−0.53	Child Routines Questionnaire	Early Adolescent Temperament Questionnaire (Parent Report)	ICQ
		Family Rituals Questionnaire		

Note. HC = household chaos; EF = executive function; CHAOS = Confusion, Hubbub, and Order Scale; DA = direct assessment; ICQ = informant-completed questionnaire.

^a Household disorganization indices: household density, visit preparedness, household cleanliness, neighborhood noise, and TV hours per day; household instability indices: number of times the child moved to another residence, changes in the primary and/or secondary caregiver, different people in the household, and household members moved into or out of the household over this period.

ment, $r = .11$, 95% CI [.05, .16], $k = 21$. The remaining moderators were not significant: household chaos dimension ($Q[2] = 1.41$, $p = .49$), sex composition ($\beta = -.01$, 95% CI [−.02, .0004]), mean child age ($\beta = .01$, 95% CI [−.004, .02]), racial/ethnic sample composition ($\beta = .001$, 95% CI [−.001, .002]), parental education ($\beta = -.001$, 95% CI [−.003, .002]), and temporal distance between household chaos and executive function measurements ($\beta = -.001$, 95% CI [−.004, .001]).

Sensitivity Analysis

As mentioned previously, for six samples there were insufficient data available to calculate effect sizes. Therefore, an effect of 0.0 was imputed in these cases and the full sample of 42 effect sizes was analyzed to estimate the impact of their inclusion. The sensitivity analysis yielded a weighted mean effect size of $r = .14$, 95% CI [.10, .18], only trivially discrepant from the mean effect size

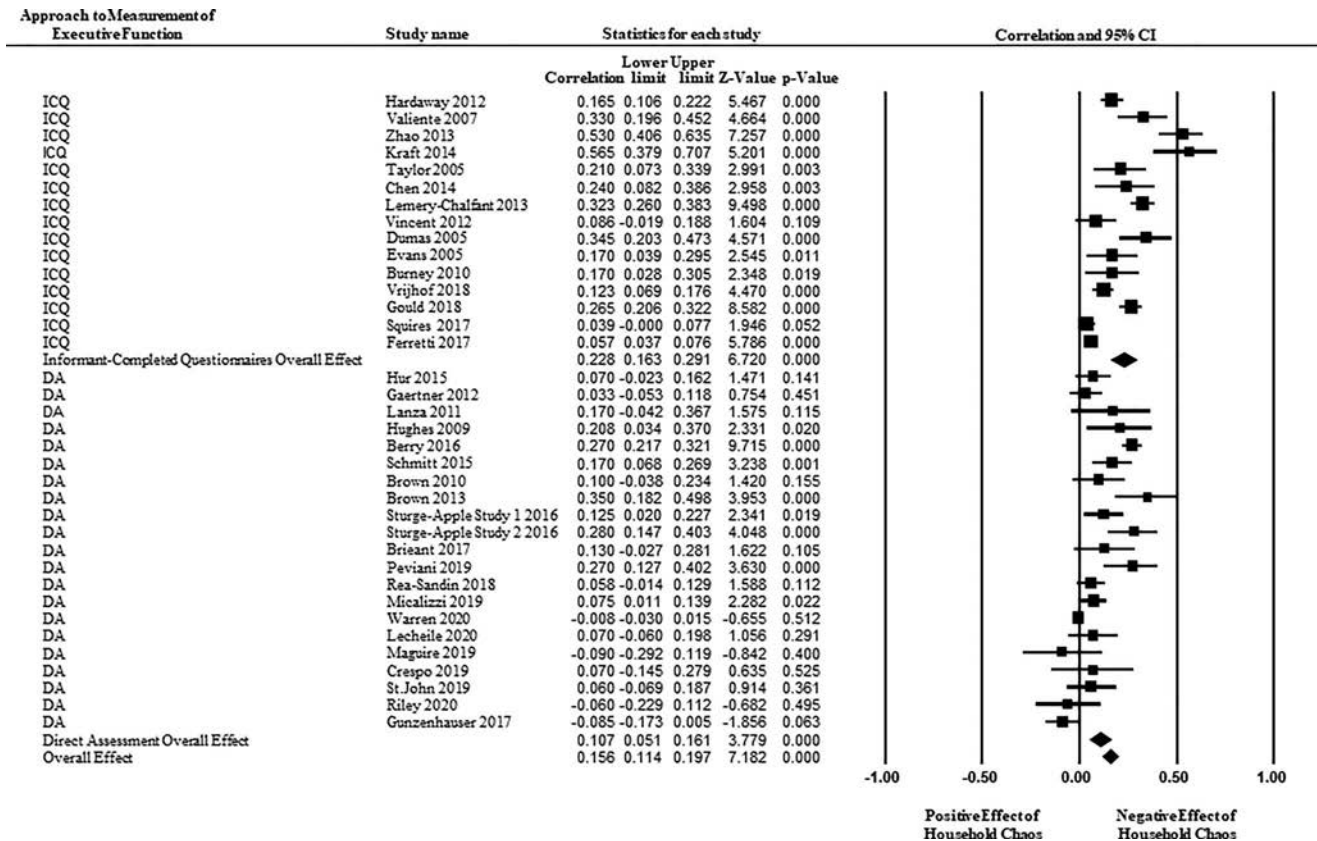


Figure 2. Forest plot with effect size estimates and confidence intervals of all studies included in meta-analysis based on measurement approach of executive functions. DA = direct assessment; ICQ = informant-completed questionnaire.

derived when these 0.0 effect sizes were excluded ($r = .16$). Heterogeneity also increased ($Q[41] = 375.79, p < .001$). Taken together, these findings indicate that the overall effect size estimate (based on $k = 36$) is robust in both magnitude and significance. Furthermore, the imputed $r = .0$ estimates fall outside the range of reported estimates and greatly increase heterogeneity, suggesting that the estimates are unrealistically conservative.

Analyses by Measurement Approach of Child Executive Functions

As highlighted in the introduction, informant-completed questionnaires and direct assessment measures share limited variance and likely tap different domains of executive function. In addition, because measurement approach to executive functions emerged as the single, significant moderator in the overall analysis, effect sizes were separated accordingly and analyzed independently. For informant-completed questionnaires, five studies previously removed from the initial overall analysis were reinserted. This had minimal impact on effect size, which diminished by .01 to yield a weighted mean strength of $r = .22, 95\% \text{ CI } [.16, .27]$. Effects were significantly heterogeneous, $Q(19) = 219.18, p < .001$, although examination of moderators—household chaos dimension ($Q[2] = 3.05, p = .22$), sex composition ($\beta = -.01, 95\% \text{ CI } [-.03, .003]$), mean child age ($\beta = .002, 95\% \text{ CI } [-.01, .02]$), racial/ethnic

sample composition ($\beta = .001, 95\% \text{ CI } [-.001, .003]$), parental education ($\beta = -.002, 95\% \text{ CI } [-.01, .002]$), temporal distance ($\beta = -.002, 95\% \text{ CI } [-.005, .002]$)—did not yield significance.

In the context of direct assessment, effects again proved significantly heterogeneous, $Q(20) = 146.45, p < .001$. Among moderators, only the chaos dimension was significant ($Q[2] = 11.56, p < .01$): instability was associated with a significantly larger effect size ($r = .17, 95\% \text{ CI } [.03, .30], k = 6$) than disorganization ($r = .06, 95\% \text{ CI } [.02, .11], k = 14$). The remaining moderators—sex composition ($\beta = -.01, 95\% \text{ CI } [-.02, .01]$), mean child age ($\beta = .003, 95\% \text{ CI } [-.01, .02]$), racial/ethnic sample composition ($\beta = .001, 95\% \text{ CI } [-.001, .003]$), parental education ($\beta = .001, 95\% \text{ CI } [-.002, .004]$), temporal distance ($\beta = .001, 95\% \text{ CI } [-.003, .004]$)—were not significant.

The Duval and Tweedie procedure did not trim any studies for any of the three meta-analyses; observed and imputed effect sizes were identical. Consequently, there is no reason to suspect publication bias in these analyses.

Sensitivity Analyses

Sensitivity analyses were conducted for both informant-completed questionnaires ($k = 24$) and direct assessment ($k = 25$) studies, which included four additional samples each, for which effect sizes were imputed as 0.0. For questionnaires, this analysis

yielded a weighted mean effect size of $r = .19$, 95% CI [.14, .24] and heterogeneity increased, $Q(23) = 246.53$, $p < .001$. The direct assessment analysis yielded a weighted mean effect size of $r = .09$, 95% CI [.04, .14] and again, heterogeneity increased, $Q(24) = 150.27$, $p < .001$. These statistics indicate that the missing data likely had little effect on the estimates of effect size (a .03 and .02 difference for questionnaires and direct assessment, respectively) and significance.

Discussion

A meta-analytic review of household chaos and executive function is important given the increasing prevalence of chaos in North American homes (Lichter & Wethington, 2010; Marsh, Dobson, & Maddison, 2020) and the vulnerability of executive function development to such environmental factors. As such, this meta-analysis provides insight into critical foci for future research. Across the 36 independent samples examined in the overall analysis of the current study, we found evidence that household chaos is significantly and negatively associated with child executive functioning (defined in its broadest terms, including both informant-completed questionnaires and direct assessments). These findings are comparable with previous meta-analyses assessing other predictors of executive functions such as SES (Lawson et al., 2018). Further, they are consistent with the bioecological theory that posits that chaos within a child's microsystem can adversely affect their development via disruption of the proximal processes with which the child is engaged (Bronfenbrenner & Evans, 2000; Wachs & Evans, 2010).

In the moderator analysis, only measurement approach of executive functions was a significant moderator with a stronger association between household chaos and child executive functions found in studies that used informant-completed questionnaires as compared with direct assessments. Informant-completed questionnaire and direct assessment effects were heterogeneous, although no moderators were identified that explained this heterogeneity in the context of questionnaires. Respecting direct assessment, effect size depended on chaos dimension (instability vs. disorganization). Further discussion of the measurement approach of executive functions and the role of household instability is provided below.

Measurement Approach of Executive Functions

Separate analyses for informant-completed questionnaires and direct assessment of executive functions were conducted for several reasons. Correlations between measures of executive functions as assessed via informant-completed questionnaires and direct assessment are poorly correlated (Anderson et al., 2002; Mahone et al., 2002; Vriezen & Pigott, 2002). Indeed, it has been argued that the two approaches measure different domains of executive function (Toplak et al., 2013) and assess differing underlying neural mechanisms (Faridi et al., 2015). Consistent with these a priori considerations, the current analysis, comprising both informant-completed questionnaires and direct assessment approaches, proved significantly heterogeneous, with assessment approach emerging as the single significant moderator.

Furthermore, it is possible that the variance contributed by measurement approach is sufficient to mask other moderator effects in the full sample. This possibility is only partially verified by

the fact that when informant-completed questionnaires and direct assessment approaches are analyzed separately, chaos dimension (disorganization vs. instability) emerges as a significant moderator in the context of direct assessment, although it is also true that the informant approach remains heterogeneous, with no significant moderators emerging.

Informant-completed questionnaires may provide an ecologically valid assessment of child executive functioning in real-world settings (Toplak et al., 2013). However, they provide global ratings that may be confounded by other abilities (e.g., emotion regulation, motivation; Snyder, Miyake, & Hankin, 2015), thus limiting their discriminant validity. Further, questionnaires are susceptible to recall inaccuracies (Joyner, Silver, & Stavinoha, 2009; Wachs, 2013), and bias (Silver, 2014). It is also important to consider the role of shared method variance when interpreting the higher effect size for informant-completed questionnaires. Of the studies that used questionnaires to measure both household chaos and child executive functions ($k = 19$), 12 involved a single, parental informant for both assessments, four involved a combination of informants (e.g., maternal report for household chaos and a composite of maternal and paternal report of child executive functions), and three had a different informant for both assessments. Although there are insufficient studies to conduct a formal analysis to assess differences across these informant groups, a preliminary examination revealed findings consistent with the hypothesis that studies with a single informant tend to yield a higher overall effect size ($r = .26$, 95% CI [.18, .34]), compared with studies with a combination of informants ($r = .20$, 95% CI [.09, .31]) and studies with two different informants ($r = .13$, 95% CI [.02, .23]). Collectively, these findings suggest the importance of multiple informants in potentially mitigating the effects of shared method variance (Duku & Vaillancourt, 2014). Furthermore, multiple informants (i.e., parents, teachers, self-reports from adolescents) may provide a range of effect sizes representing a more nuanced, cross-situational assessment of executive functions. For example, low agreement in parent and teacher ratings of child outcomes has been demonstrated (Achenbach, McConaughy, & Howell, 1987; Duku & Vaillancourt, 2014) and is likely attributable to differences in context within which the informant observes the child's behavior. Children may behave differently across settings due to variations in structure, demand, or expectations (De Los Reyes & Kazdin, 2005). Moreover, in the context of household chaos, it is plausible that parents within the same environment as their children would also be subject to the effects of chaos, thereby potentially negatively skewing their perspective of their child's behaviors. Therefore, although inclusion of multiple informants can be challenging as has been noted in the child psychopathology literature, the use of multiple informants can strengthen the assessment of child outcomes if proper theoretical models inform the selection of informants (Kraemer et al., 2003) as well as statistical modeling, such as structural equation modeling (SEM; Martel, Markon, & Smith, 2017). Taken together, meaningful information can be obtained via inclusion of multiple informants and in examining the variability in informant ratings across contexts (Dirks, De Los Reyes, Briggs-Gowan, Cella, & Wakschlag, 2012).

Direct assessments may provide a more structured, refined setting to detect specific executive function skills due to their focus on how one processes information (e.g., response time, accuracy; Toplak et al., 2013). However, its estimation may be over-

zen & Wilhelm, 1996) or underestimated (Cripe, 1996; Sbordone, 1996) given its rigid approach. Moreover, its highly structured approach limits a child's ability to use compensatory skills that they may otherwise use in real world contexts (Chaytor, Schmitter-Edgecombe, & Burr, 2006). With that said, in the current review, direct assessments demonstrated some sensitivity to moderating factors, despite their smaller effect size. This underscores the possible important role of direct assessment measures to a nuanced understanding of the relation between chaos and executive function. Ultimately, more research is needed, and it may be that a multidimensional approach incorporating both questionnaire and direct assessment methodologies is necessary to the comprehensive study of chaos and executive function.

Moderation Within Direct Assessments

The instability dimension of household chaos was more strongly associated with child executive function than the disorganization dimension. Although not considered to be as prevalent as disorganization, even some exposure to instability during early and middle childhood (e.g., changes in residences, housing structure and/or caregiver structure) can have significant effects on behavioral functioning (Fomby & Cherlin, 2007; Tiesler et al., 2013; Ziolk-Guest & McKenna, 2014). It is possible that the unpredictability within the home minimizes a child's control over their environment and instills a sense of helplessness (Evans & Stecker, 2004), and without proper support (e.g., from a caregiver) this could hinder the child's ability to manage their affect and behaviors, which have been linked to underlying deficits of executive functions (Sulik et al., 2015). For example, a 6-year longitudinal study showed higher self-regulation, less alcohol use, lower epinephrine levels, and greater rates of entering postsecondary education in youths who were exposed to more routines (i.e., eating meals as a family, consistent bedtime; Barton et al., 2019). Further, difficulties in executive functioning have been reported in children who moved at least once between preschool and fifth grade (Roy, McCoy, & Raver, 2014), as well as those that experienced changes to maternal partners in coresidential relationships (Cooper, Osborne, Beck, & McLanahan, 2011).

Although instability seems to have a potent effect on child executive functions, given the small dataset, it would be important to further investigate these associations within a larger scale, longitudinal study. It is possible that the higher effect size for household instability as compared to household disorganization may be due to other factors such as the specific type of executive function being assessed, or the number of executive functions being studied. When assessed preliminarily in the current meta-analysis, neither of these factors significantly interacted with the household chaos dimension. However, this could be attributable to reduced sample sizes common with moderation analyses resulting from missing data or restricted range (i.e., majority of the outcomes were inhibition or effortful control). The effect of age on the association between household chaos dimension and executive functioning was also assessed to examine whether instability may have a greater adverse effect on older rather than younger children. However, this interaction was also not significant. Future studies should aim to further explore such factors that may interact with chaos dimension and exacerbate adverse effects on child executive functions. Ultimately, there is a need to better represent the insta-

bility dimension of household chaos in its study, given an apparent gap in examining this dimension of chaos with respect to its differential and joint (with disorganization) effects on child executive functioning.

Potential Mechanisms of Effect

According to the bioecological theory, household chaos may disrupt the sustainability and predictability of proximal processes, which can have adverse consequences for child outcomes (Bronfenbrenner & Evans, 2000; Wachs & Evans, 2010). This may occur by increasing stress in children, which in turn may disrupt their executive function development. Interestingly, in one study, confirmatory factor analyses conducted with two independent samples demonstrated that measures of household instability and stressful life events, although positively correlated, loaded onto distinct factors (Marcynyszyn, Evans, & Eckenrode, 2008). Instability impacted youth adjustment above and beyond stressful negative life events. These findings suggest that consistency and predictability are empirically distinct from stressful life events and exploration into their independent and shared contributions to child outcomes is warranted in future studies. To that point, exposure to chronic stress leads to dysregulation of the hypothalamic-pituitary-adrenal (HPA) axis which can result in potentially lasting structural and functional changes in the prefrontal cortex and associated regions (e.g., hippocampus, amygdala; Arnsten, 2009; Teicher et al., 2003)—all areas of the brain relevant to executive function, memory and emotion regulation. Chronic stress, then, is a likely mechanism through which household chaos exerts its effects on child executive functioning and its underlying neurobiology and should be explored in future research.

A second possible mechanism is a *system of chaos*, which may be generated via disruptions to parent-child interactions—a central proximal process in early childhood. The interaction between parent and child is reliant, in part, on the executive functioning of each party. If both parent and child have deficits in their executive functioning, subsequent dyadic interactions may then be more disruptive, generating an overall system of chaos. Within this system of chaos parenting behaviors can also be disrupted. For example, higher chaos within the home has been linked to less responsiveness, sensitivity, and involvement of parents with their children (Wachs, 2005), less warmth and greater exposure to stressful events (Deater-Deckard et al., 2009), lower maternal emotional availability (Whitesell, Teti, Crosby, & Kim, 2015), and reports of inconsistent discipline (Dumas et al., 2005). The quality of parent-child interactions and its effect on child executive functioning is extensively documented (Bernier, Carlson, Deschênes, & Matte-Gagné, 2012; Hughes & Devine, 2019). However, few studies have examined the association between household chaos, parenting, and child executive functions (Valiente, Lemery-Chalfant, & Reiser, 2007; Vernon-Feagans et al., 2016). Specifically, parental positive reactions to child emotions significantly mediated the association between household chaos and child effortful control (Valiente et al., 2007). Additionally, cumulative household disorganization exerted a negative, indirect effect on the behavioral regulation of kindergarten-aged children via parenting acceptance and responsivity and early executive function skills (Vernon-Feagans et al., 2016). Importantly, attachment security may also be disrupted by household chaos—given its association

to parenting behaviors—which would presumably adversely affect child executive functions. However, these associations have yet to be examined. Taken together, it is possible that (a) parental executive function deficits may impact the dyadic interaction with the child creating a system of chaos and (b) the reciprocal relationship between chaos and parental behaviors may further exacerbate negative effects on child executive functioning.

Limitations

This meta-analysis represents a comprehensive review of all identified studies of household chaos published to date. Importantly, it showed a substantial and significant effect size difference when informant-completed questionnaires and direct assessments were compared. Of note, the high precision of the estimates was consistent across all three analyses indicative of the low variability of the estimates and adding to the credence of the results (Brand & Bradley, 2016). We considered maintaining a combined (informant-completed questionnaire + direct assessment) approach and simply evaluating all potential moderators in terms of their interaction with methodological approach. However, it appears that the two approaches differ in their focus sufficiently to preclude their common analysis. Furthermore, as with all meta-analyses, sample size diminishes in the context of moderation analysis because relevant data are often missing. For example, we were unable to assess the impact of child verbal ability, parental age, and certain SES indices (e.g., income-to-needs ratio) due to missing data and/or restricted data ranges. This signals the need for future researchers to include relevant data for the purpose of analyzing important moderators. Relatedly, a power analysis was not conducted to determine whether there was sufficient statistical power to accurately detect moderating effects despite the current study including 36 data sets and an overall sample size of 16,480. As per recommended best practices, and because of other factors (beyond sample size) that can affect statistical power (Aguinis, Pierce, Bosco, Dalton, & Dalton, 2011), future meta-analyses should incorporate an a priori power analysis (Aguinis, Gottfredson, & Wright, 2011). Another limitation of the current meta-analysis was that the data included were largely correlational. Although ethical considerations preclude experimental manipulation of environmental features, quasi-experimental designs are necessary to support causal inferences from household chaos to child executive functions. Further, more studies with longitudinal designs are needed and would permit (a) exploration of within-person fixed effects to eliminate potential unobserved, time-invariant confounds and (b) examination of possible interaction effects between household chaos dimension and child age to provide insight into whether certain developmental periods are more vulnerable to one or both dimensions. Further to this latter point, future studies also need to explore the potential differential effects of the dimensions of household chaos. Most studies focus on the total score of the CHAOS scale without parsing out the items relevant to each dimension. It would be advisable for future studies to consider quantifying the percentage of each dimension represented in the measure of household chaos used; thereby providing a more precise evaluation and mitigating any confounding variance. Another limitation may be the inclusion of effortful control and executive function in a single construct. Although some studies support this integration (Tiego et al., 2020; Zhou et al., 2012), further examination into the most accurate conceptualization of the construct of executive function, particularly in early childhood, is

warranted. Finally, in six cases, data linking chaos and executive functions were missing. We excluded these studies from relevant analyses (which, in theory, could inflate estimated effect size based on publication bias); however, sensitivity analyses wherein effect sizes were imputed as zero revealed only trivial impact.

Conclusion

Household chaos poses a significant risk to the development of executive functions in children. The current meta-analysis highlights the importance of multimethod approaches in measuring executive function in children to ensure rigorous and accurate assessment. These findings can inform the development of preventative strategies for children and their families aimed at the reduction of disorganization (e.g., minimizing clutter and ambient noise) and providing greater stability (e.g., promotion of routines) within the home environment. There are evidence-based interventions for executive functioning; however, a more multifaceted approach may be required, targeting household chaos as a risk factor and child executive function as an outcome to tailor to interventions to the specific needs of families.

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- References marked with an asterisk indicate studies included in the meta-analysis.
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